

Parameter recovery

$y \sim 1 + (1 \mid \text{subject_id}) + (1 \mid \text{session_id}) + (1 \mid \text{subject_id}:\text{session_id})$

500

agents

2

sessions

1

trials per session

R-hat 1.051

1170 divergences

Design

Data were generated from 500 agents, 2 sessions, and 1 observations per agent per session. The intercept was fixed at 0. Agent-level random effects were drawn as $\text{gamma_subject} \sim \text{Normal}(0, 1)$, $\text{gamma_session} \sim \text{Normal}(0, 0.5)$, and $\text{gamma_subjectXsession} \sim \text{Normal}(0, 0.5)$. Residual error was $\text{Normal}(0, 0.5)$.

ICC_agreement and ICC_consistency were stored as true values (0.57 and 0.67) using the two-way single-measure formulas on subject-session means (variances, with subject-by-session interaction and trial error / n_trials in the residual); they were not used to generate data and were recovered from the scale parameters. No parameter was transformed onto the unit interval. Parameters were recovered with brms. The sampler used 4 chains, 5000 iterations, and 3000 warmup.

Population recovery

Parameter	True	Median	5%	95%	Covered
intercept	0.000	-0.200	-2.256	1.806	yes
sigma_error	0.500	0.615	0.391	0.712	yes
sigma_subject	1.000	1.028	0.964	1.096	yes
sigma_session	0.500	1.482	0.356	5.450	yes
sigma_subjectXsession	0.500	0.328	0.032	0.581	yes
ICC_agreement	0.571	0.283	0.035	0.635	yes
ICC_consistency	0.667	0.684	0.644	0.722	yes

Agent-level recovery

Parameter	r	Bias	Slope	Range	Post. SD	Pass r	Pass bias	Pass prec.
gamma_subject	0.906	0.049	0.830	1.021	0.445	yes	yes	yes
gamma_subjectXsession	0.529	0.001	0.157	0.263	0.322	no	yes	no

Figure 1. Population location

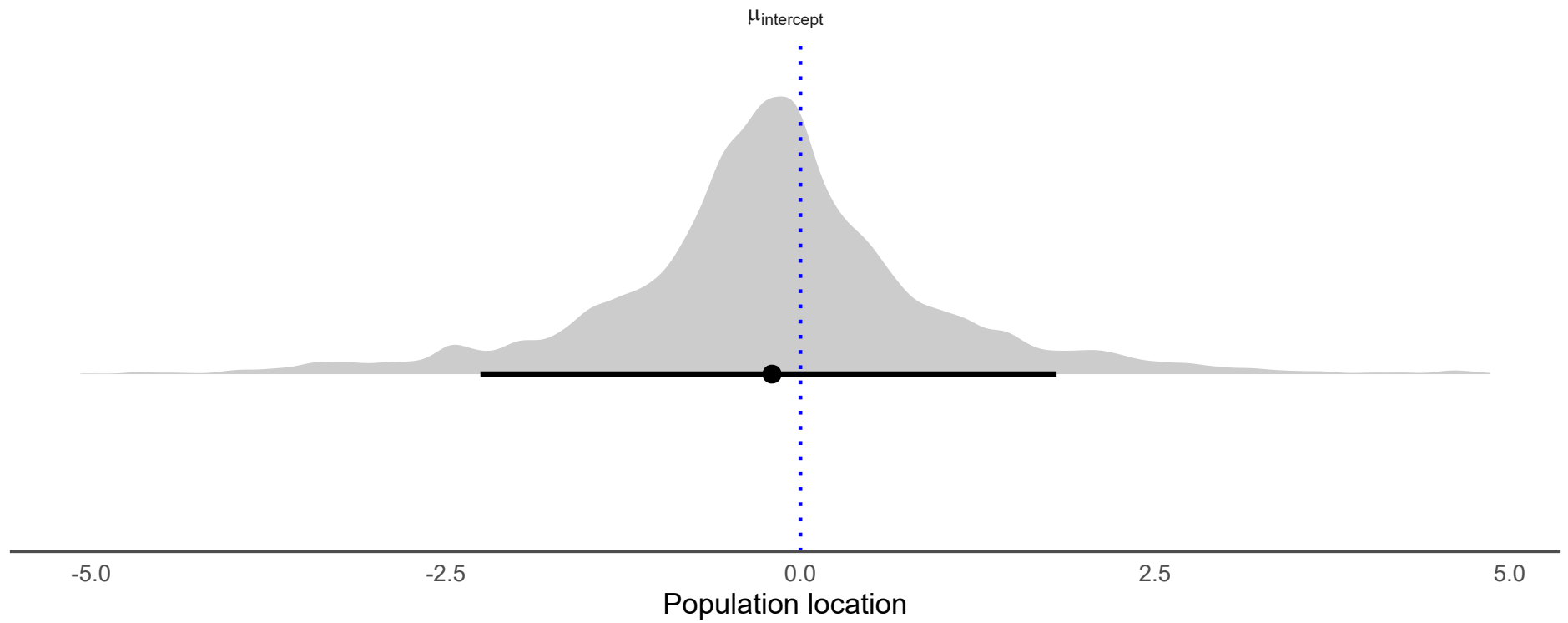


Figure 1. Posterior density of the population location (intercept). The dotted blue line marks the true generating value.

Figure 2. Population scale

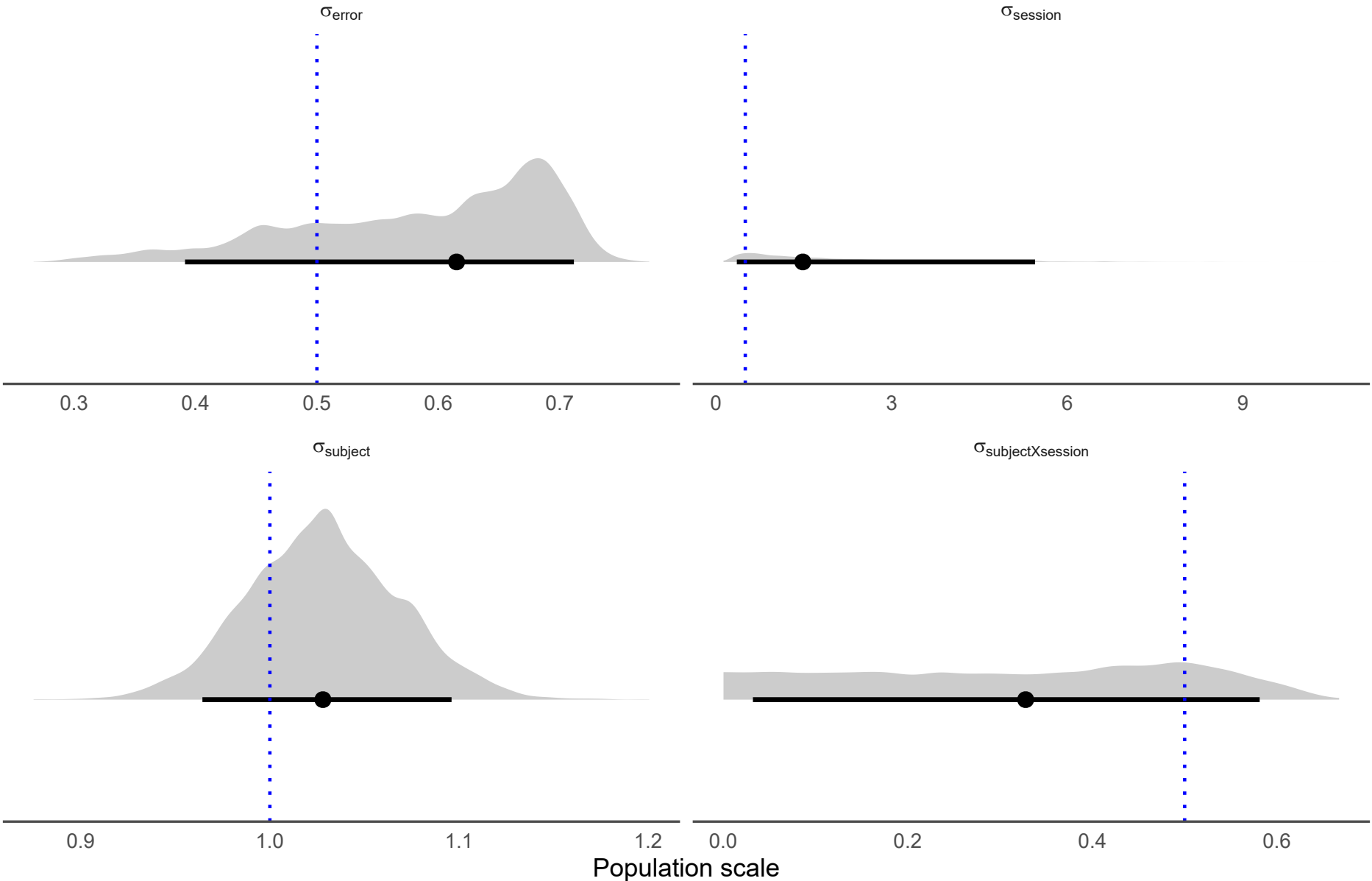


Figure 2. Posterior densities of the population scale parameters. The dotted blue line marks the true generating value on each panel.

Figure 3. Intraclass correlation

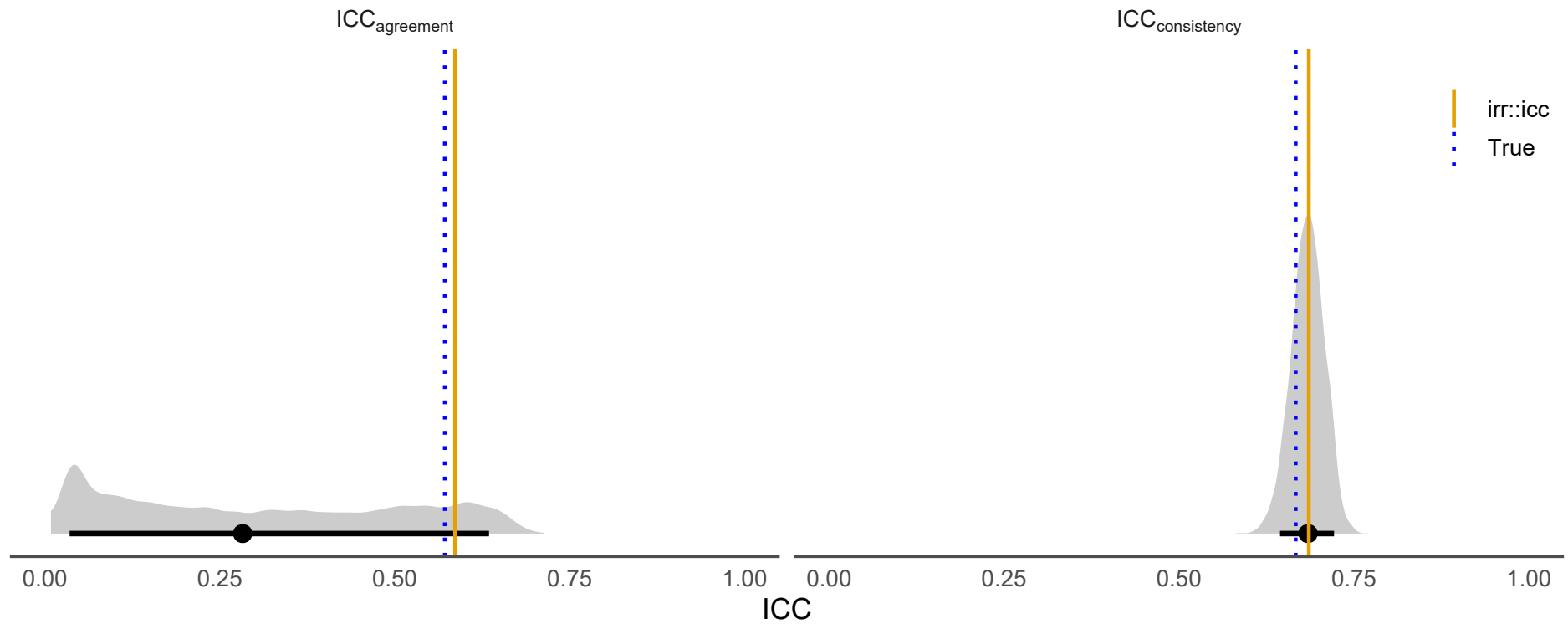


Figure 3. Posterior densities of $ICC_{agreement}$ and $ICC_{consistency}$, derived from the scale-parameter draws. The dotted blue line marks the true value stored with the generating parameters. The solid orange line marks the `irr::icc` estimate from subject-session means (twoway, single unit; agreement or consistency matching the panel).

Figure 4. Agent-level recovery

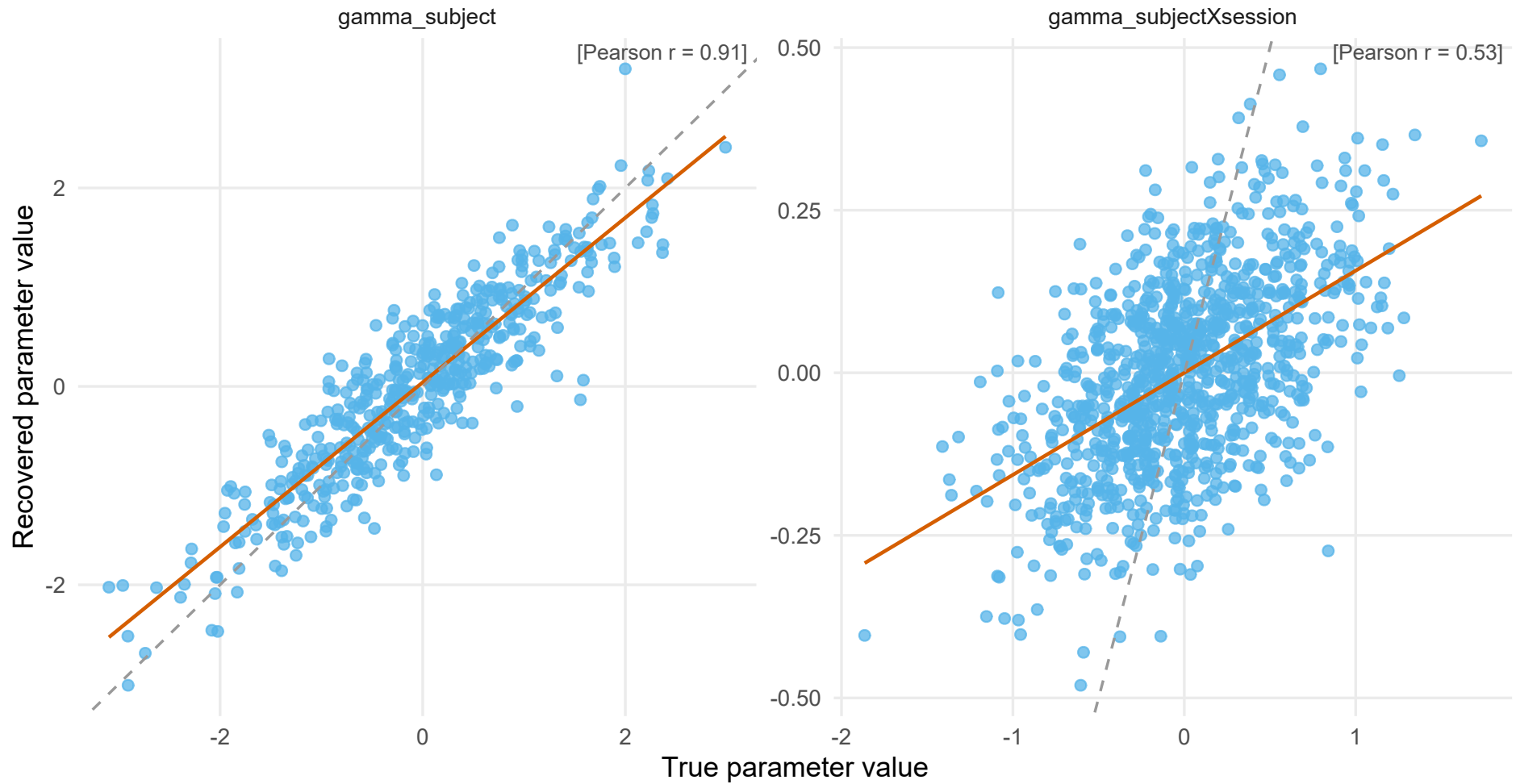


Figure 4. True versus recovered agent-level parameters. The dashed line is the identity; the solid line is the OLS fit; r is Pearson's correlation.